



Electronic I

Lecture 3

Diode Applications : Series and Parallel Configurations, Half and Full-Wave Rectification, Clippers, and Clampers

Assistant Lecturer. Saif Almansouri

Al-Nahrain University / College of Engineering

Department of Computer Engineering

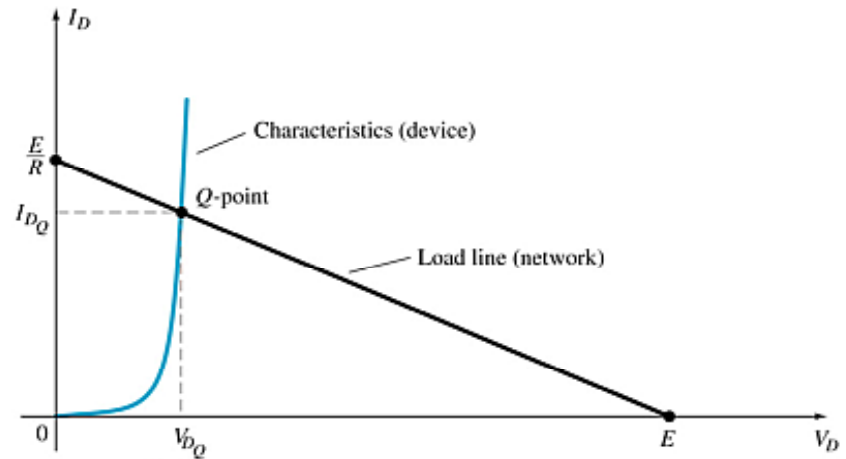
First Year / Spring Semester

safe.abduljassim@eng.nahrainuniv.edu.iq

Load-Line Analysis

The load line plots all possible combinations of diode current (I_D) and voltage (V_D) for a given circuit. The maximum I_D equals E/R , and the maximum V_D equals E .

The point where the load line and the characteristic curve intersect is the Q-point, which identifies I_D and V_D for a particular diode in a given circuit.



Series Diode Configurations

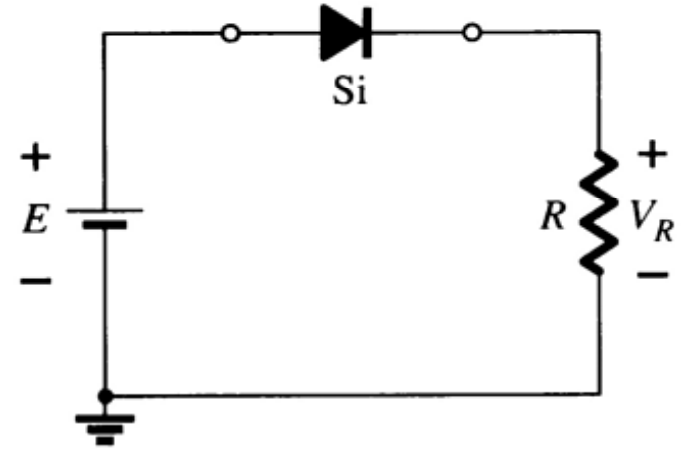
Forward Bias

Constants

- Silicon Diode: $V_D = 0.7 \text{ V}$
- Germanium Diode: $V_D = 0.3 \text{ V}$

Analysis (for silicon)

- $V_D = 0.7 \text{ V}$ (or $V_D = E$ if $E < 0.7 \text{ V}$)
- $V_R = E - V_D$
- $I_D = I_R = I_T = V_R / R$



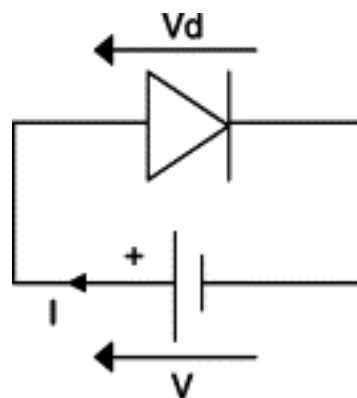
Series Diode Configurations

Reverse Bias

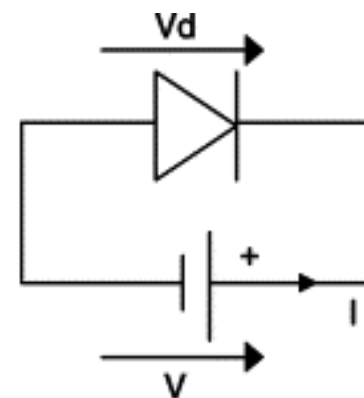
Diodes ideally behave as open circuits

Analysis

- $V_D = E$
- $V_R = 0 \text{ V}$
- $I_D = 0 \text{ A}$



Forward biased

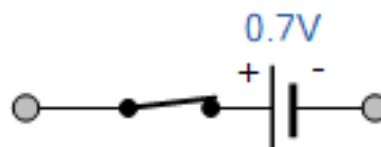


Reverse biased

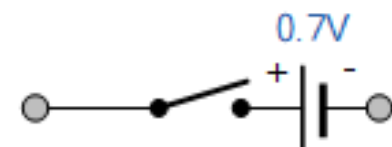
Forward Biased



Reversed Biased



Forward Bias
(switch closed)

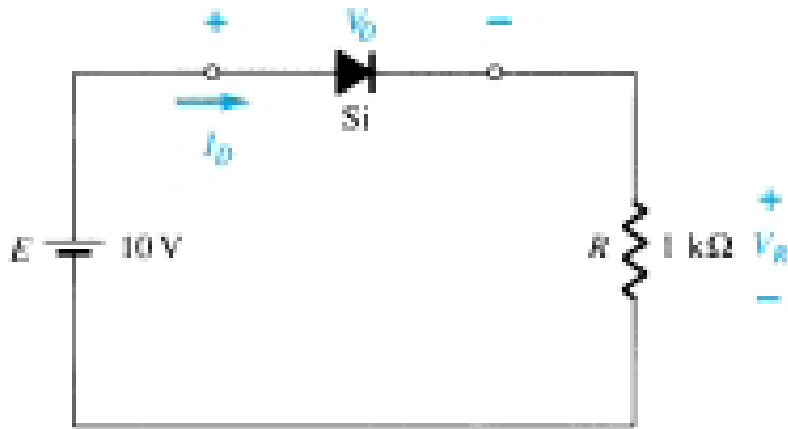


Reverse Bias
(switch open)

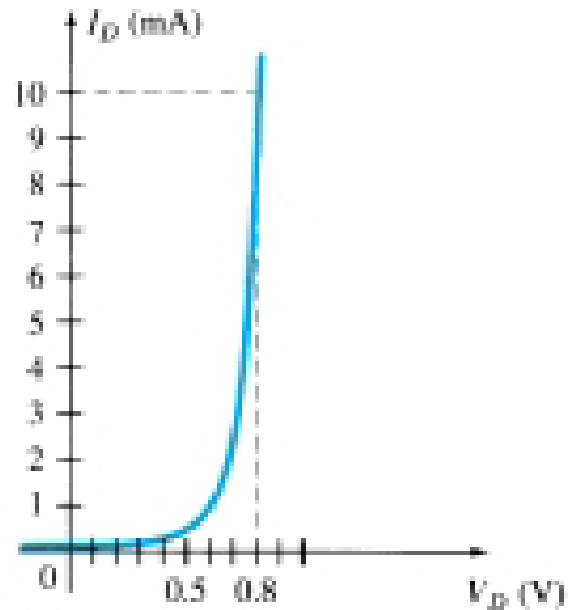
EXAMPLE

For the series diode determine

(a) V_{DQ} and I_{DQ} .



(a)



(b)

$$(a) \ I_D = \left. \frac{E}{R} \right|_{V_D=0 \text{ V}} = \frac{10 \text{ V}}{(1 \text{ k}\Omega)} = 10 \text{ mA}$$

$$V_D = E|_{I_D=0 \text{ A}} = 10 \text{ V}$$

The intersection between the load line and the characteristic curve defines the Q -point as

$$V_{D_Q} \cong \mathbf{0.78 \text{ V}}$$

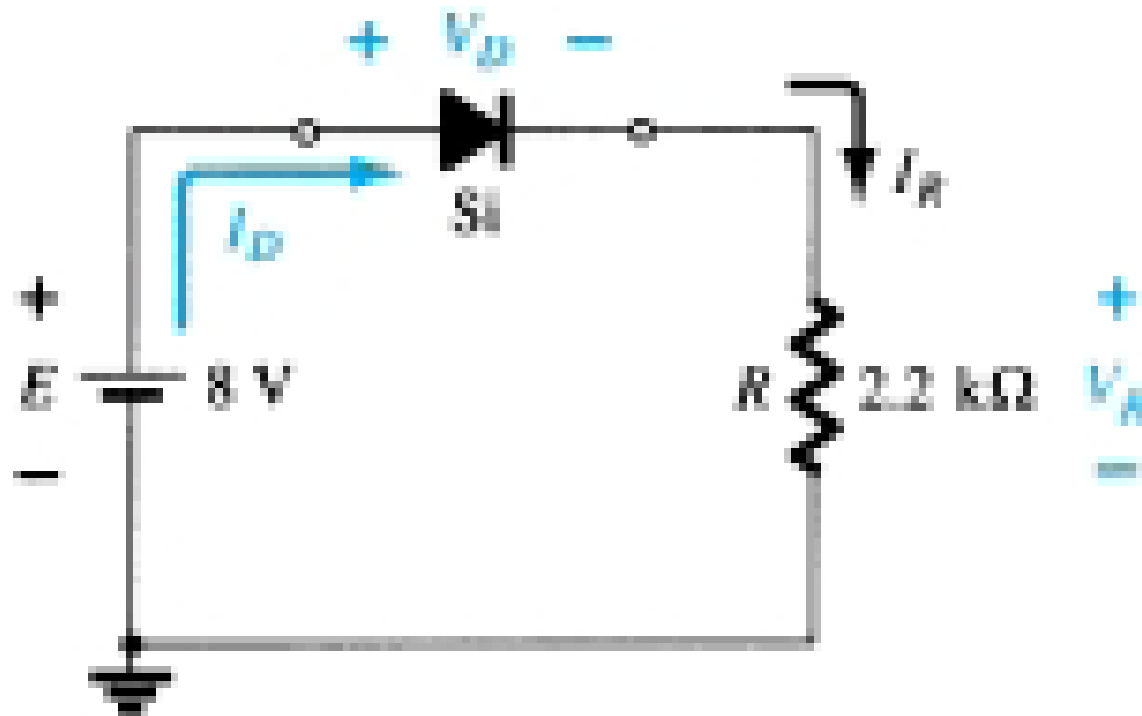
$$I_{D_Q} \cong \mathbf{9.25 \text{ mA}}$$

$$(b) \ V_R = I_R R = I_{D_Q} R = (9.25 \text{ mA})(1 \text{ k}\Omega) = \mathbf{9.25 \text{ V}}$$

$$\text{or } V_R = E - V_D = 10 \text{ V} - 0.78 \text{ V} = \mathbf{9.22 \text{ V}}$$

EXAMPLE

For the series diode configuration, determine V_D , V_R , and I_D .



- Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the “on” state,

$$V_D = \mathbf{0.7\ V}$$

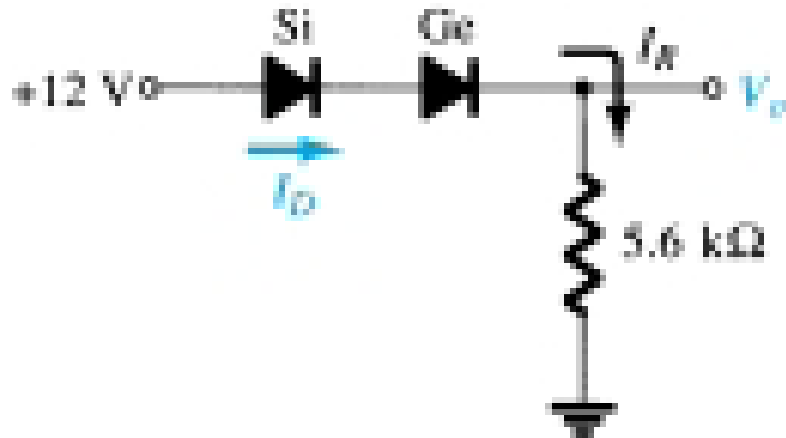
$$V_R = E - V_D = 8\ \text{V} - 0.7\ \text{V} = \mathbf{7.3\ V}$$

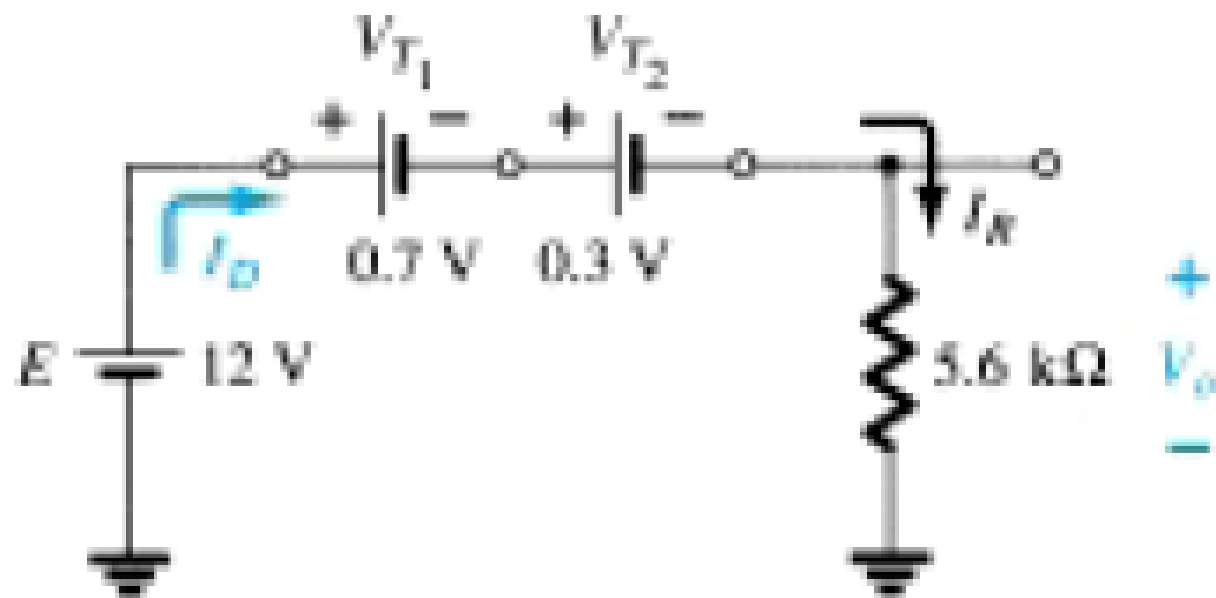
$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\ \text{V}}{2.2\ \text{k}\Omega} \cong \mathbf{3.32\ \text{mA}}$$

Series Configurations

EXAMPLE

- Determine V_o and I_D for the series circuit





$$V_o = E - V_{T_1} - V_{T_2} = 12\text{ V} - 0.7\text{ V} - 0.3\text{ V} = \mathbf{11\text{ V}}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{V_o}{R} = \frac{11\text{ V}}{5.6\text{ k}\Omega} \cong \mathbf{1.96\text{ mA}}$$

Parallel Configurations

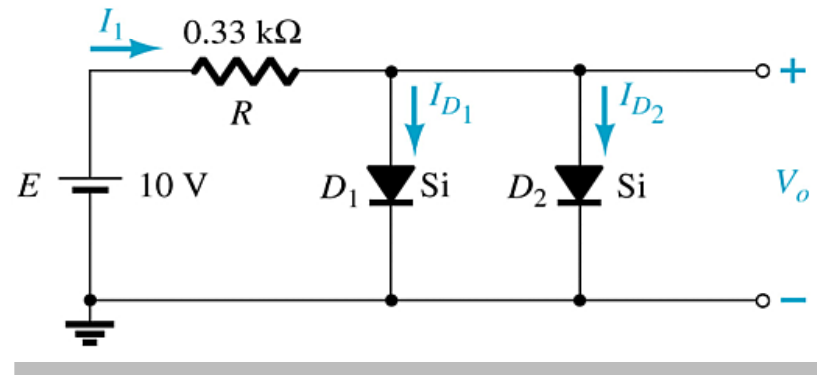
$$V_D = 0.7 \text{ V}$$

$$V_{D1} = V_{D2} = V_O = 0.7 \text{ V}$$

$$V_R = 9.3 \text{ V}$$

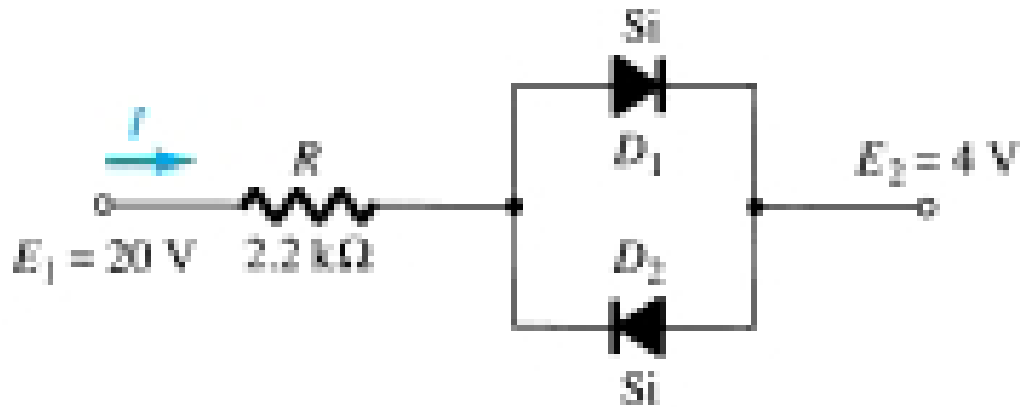
$$I_R = \frac{E - V_D}{R} = \frac{10 \text{ V} - .7 \text{ V}}{.33 \text{ k}\Omega} = 28 \text{ mA}$$

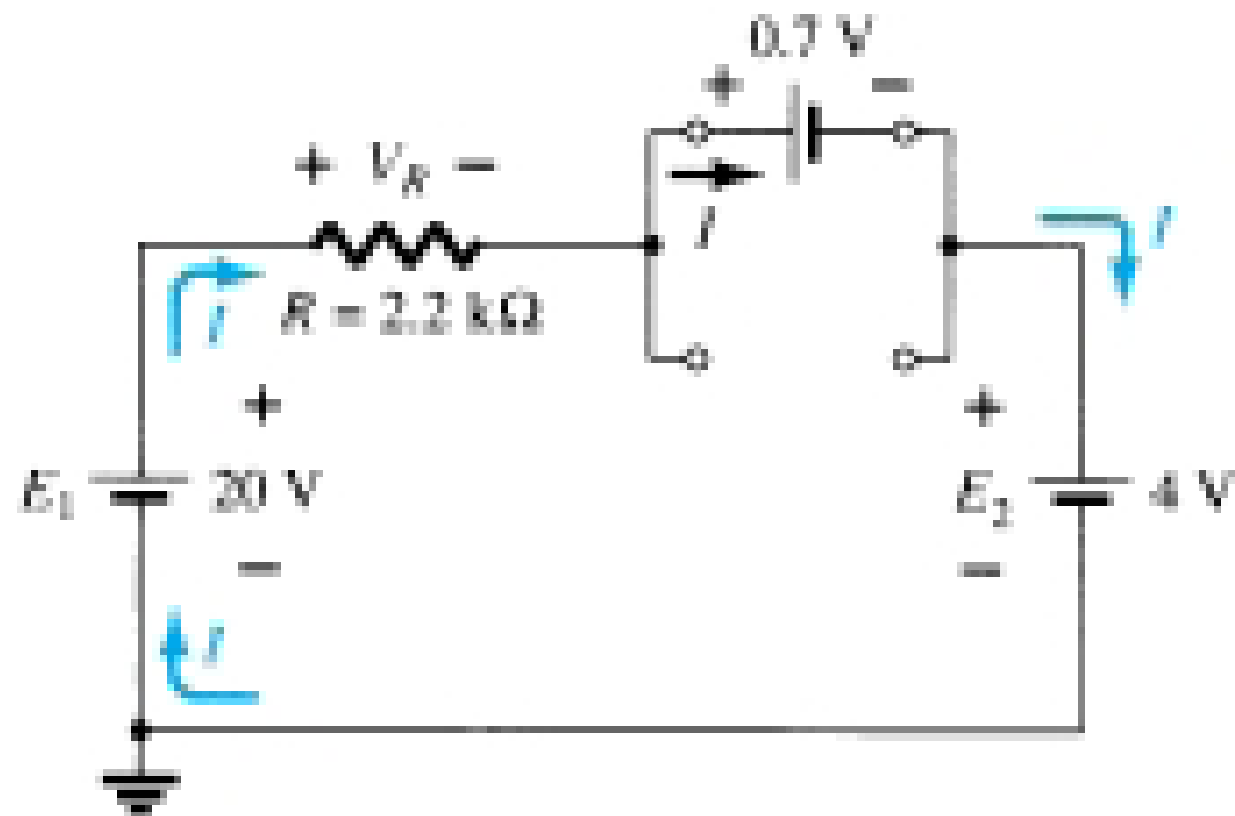
$$I_{D1} = I_{D2} = \frac{28 \text{ mA}}{2} = 14 \text{ mA}$$



Example

Determine the current I for the network:

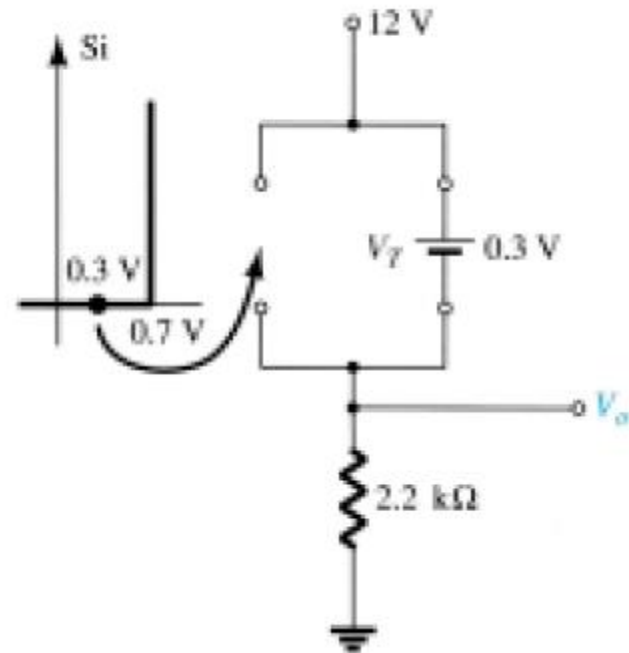
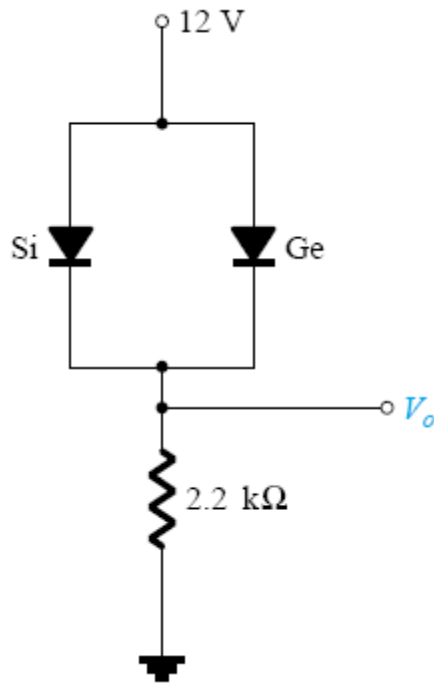




$$I = \frac{E_1 - E_2 - V_D}{R} = \frac{20\text{ V} - 4\text{ V} - 0.7\text{ V}}{2.2\text{ k}\Omega} \cong 6.95\text{ mA}$$

Example:

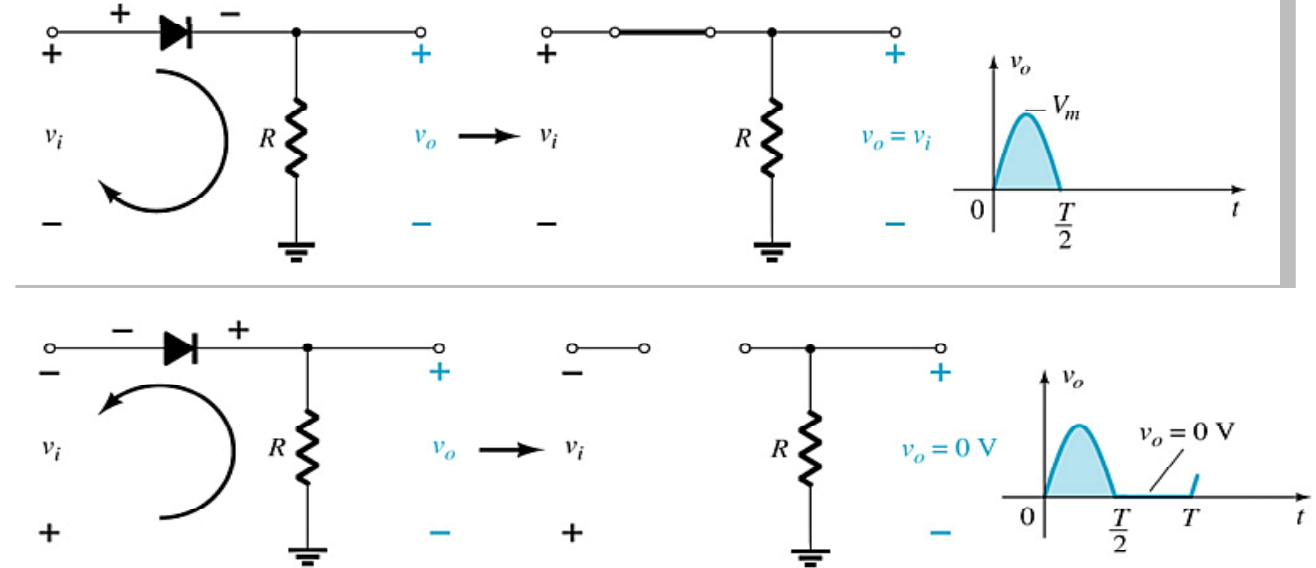
Determine the voltage V_o for the network



$$V_o = 12 \text{ V} - 0.3 \text{ V} = 11.7 \text{ V}$$

Half-Wave Rectification

The diode only conducts when it is forward biased, therefore only half of the AC cycle passes through the diode to the output.



The DC output voltage is $0.318V_m$, where V_m = the peak AC voltage.

PIV (PRV)

Because the diode is only forward biased for one-half of the AC cycle, it is also reverse biased for one-half cycle.

It is important that the reverse breakdown voltage rating of the diode be high enough to withstand the peak, reverse-biasing AC voltage.

$$\text{PIV (or PRV)} > V_m$$

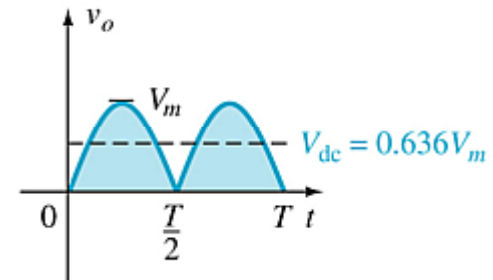
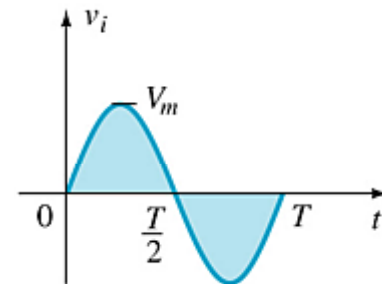
- **PIV = Peak inverse voltage**
- **PRV = Peak reverse voltage**
- **V_m = Peak AC voltage**

Full-Wave Rectification

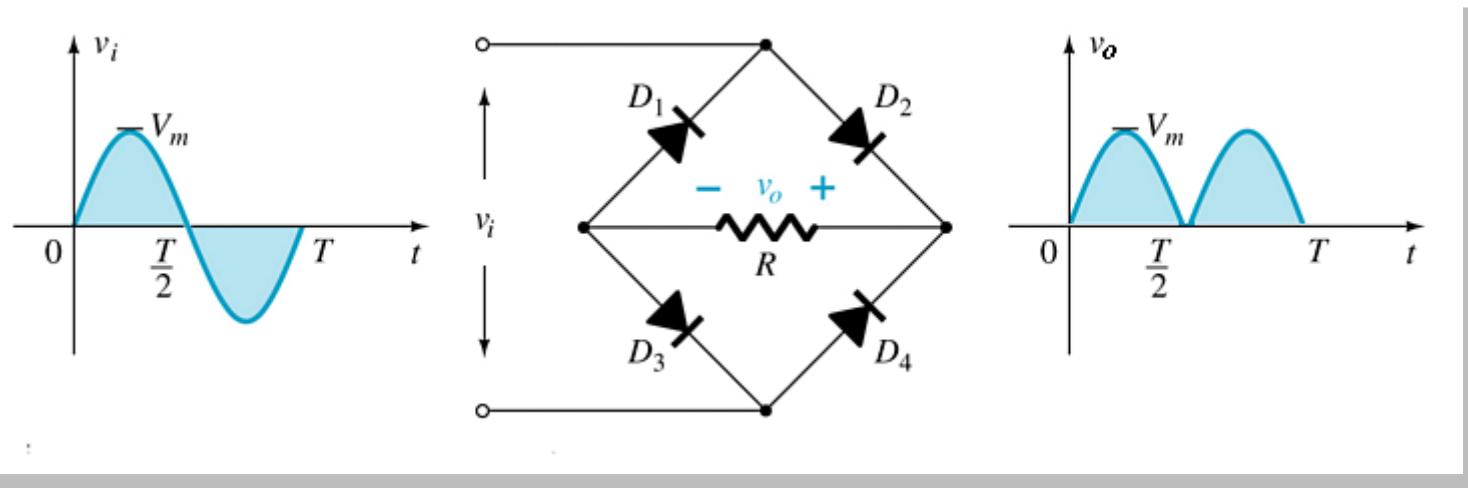
The rectification process can be improved by using a full-wave rectifier circuit.

Full-wave rectification produces a greater DC output:

- Half-wave: $V_{dc} = 0.318V_m$
- Full-wave: $V_{dc} = 0.636V_m$

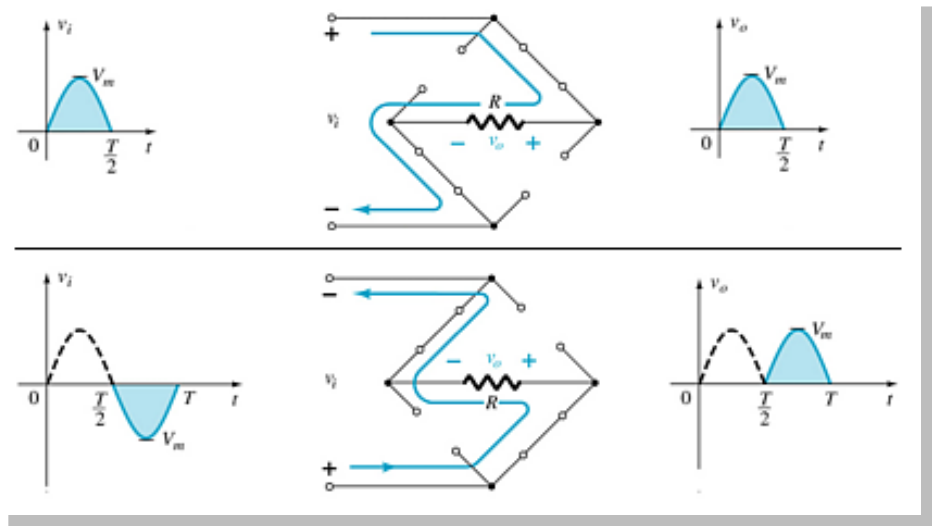


Full-Wave Rectification

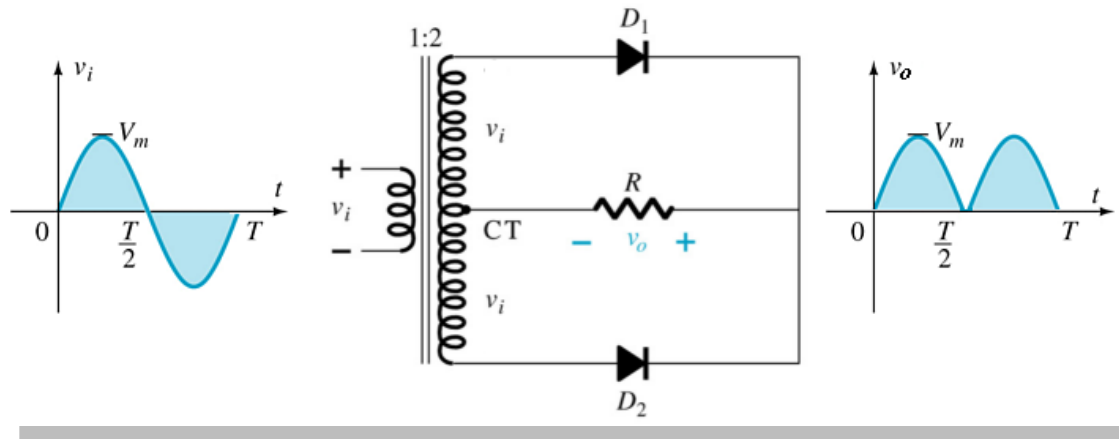


Bridge Rectifier

- Four diodes are connected in a bridge configuration
- $V_{DC} = 0.636V_m$



Full-Wave Rectification

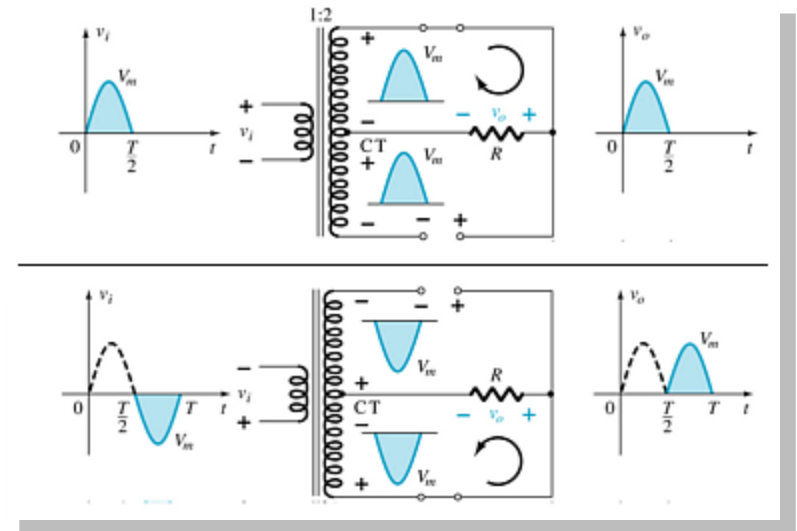


Center-Tapped Transformer Rectifier

Requires

- Two diodes
- Center-tapped transformer

$$V_{DC} = 0.636V_m$$



Summary of Rectifier Circuits

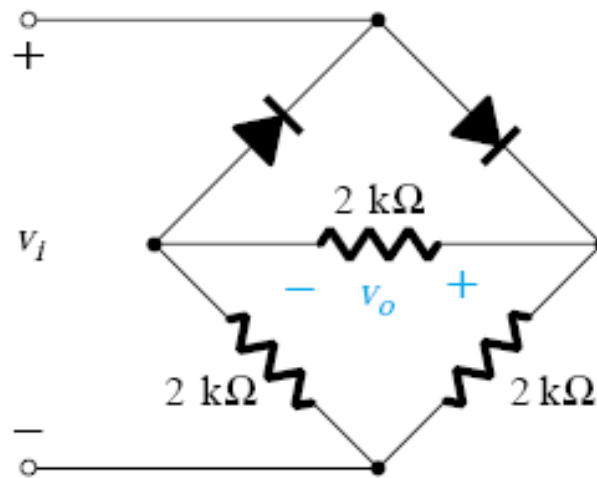
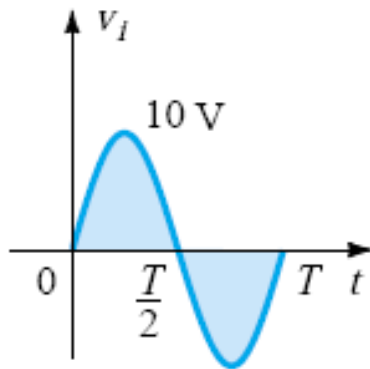
Rectifier	Ideal V_{DC}	Realistic V_{DC}
Half Wave Rectifier	$V_{DC} = 0.318V_m$	$V_{DC} = 0.318V_m - 0.7$
Bridge Rectifier	$V_{DC} = 0.636V_m$	$V_{DC} = 0.636V_m - 2(0.7 \text{ V})$
Center-Tapped Transformer Rectifier	$V_{DC} = 0.636V_m$	$V_{DC} = 0.636V_m - 0.7 \text{ V}$

V_m = peak of the AC voltage.

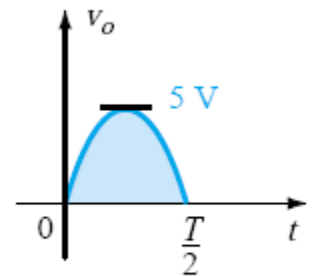
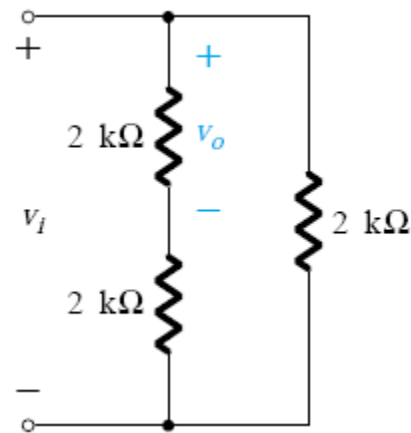
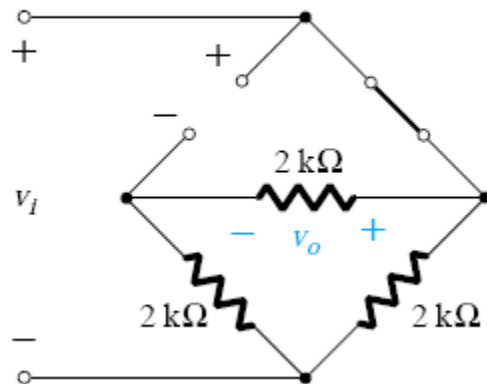
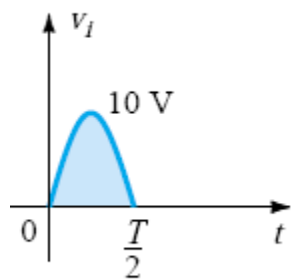
In the center tapped transformer rectifier circuit, the peak AC voltage is the transformer secondary voltage to the tap.

EXAMPLE

Determine the output waveform for the following network and calculate the output dc level and the required PIV of each diode.



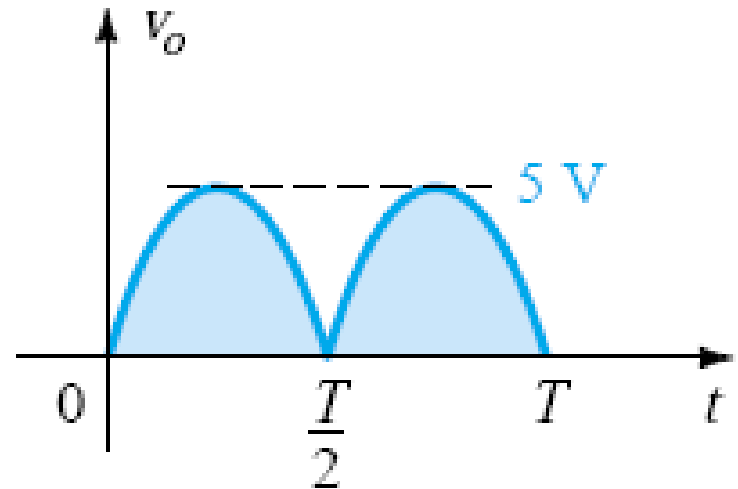
For the positive region of the input voltage



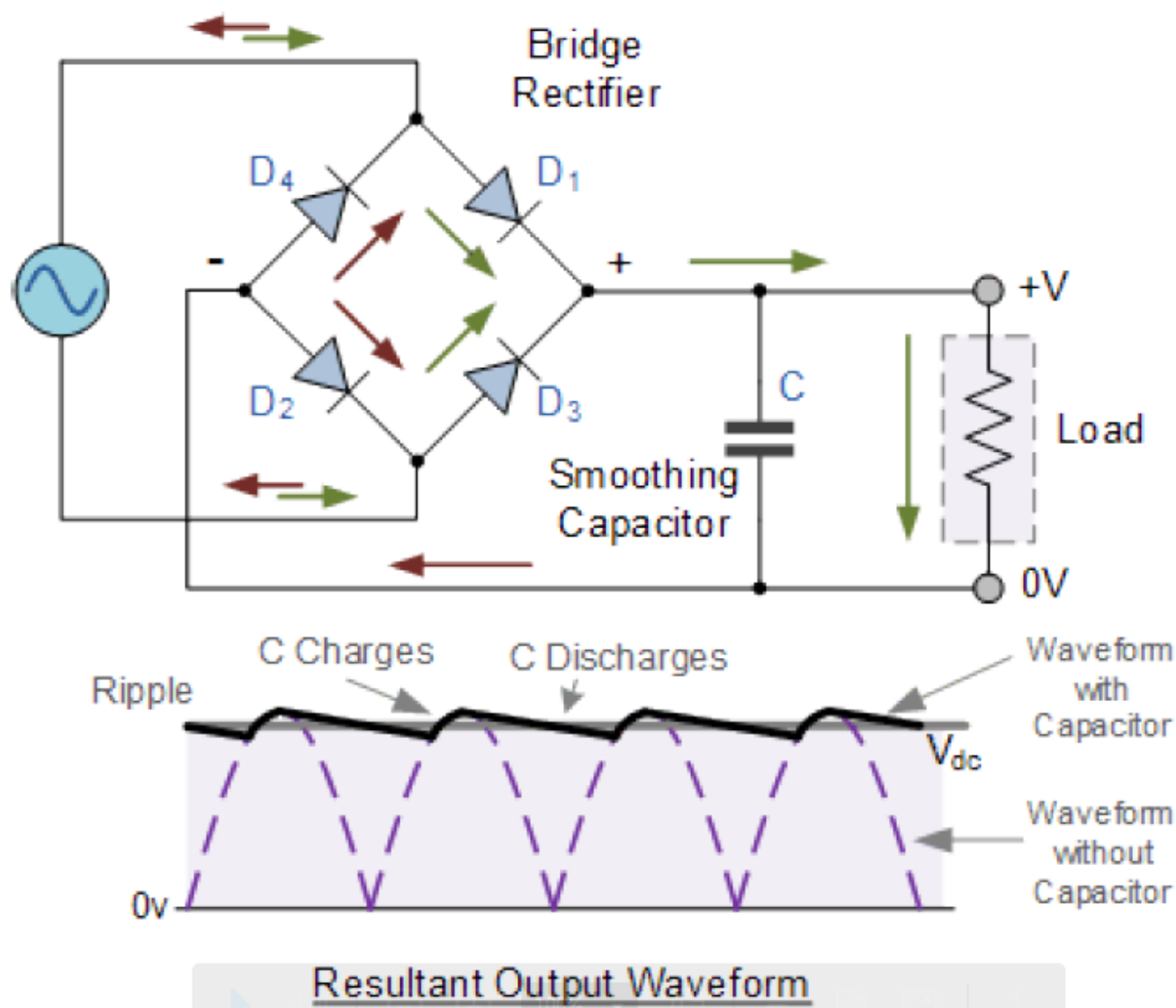
$$v_o = \frac{1}{2} v_I \text{ or } V_{o_{\max}} = \frac{1}{2} V_{I_{\max}} = \frac{1}{2}(10 \text{ V}) = 5 \text{ V},$$

For the negative part of the input the roles of the diodes will be interchanged and v_o will appear as

$$V_{dc} = 0.636(5 \text{ V}) = \mathbf{3.18 \text{ V}}$$



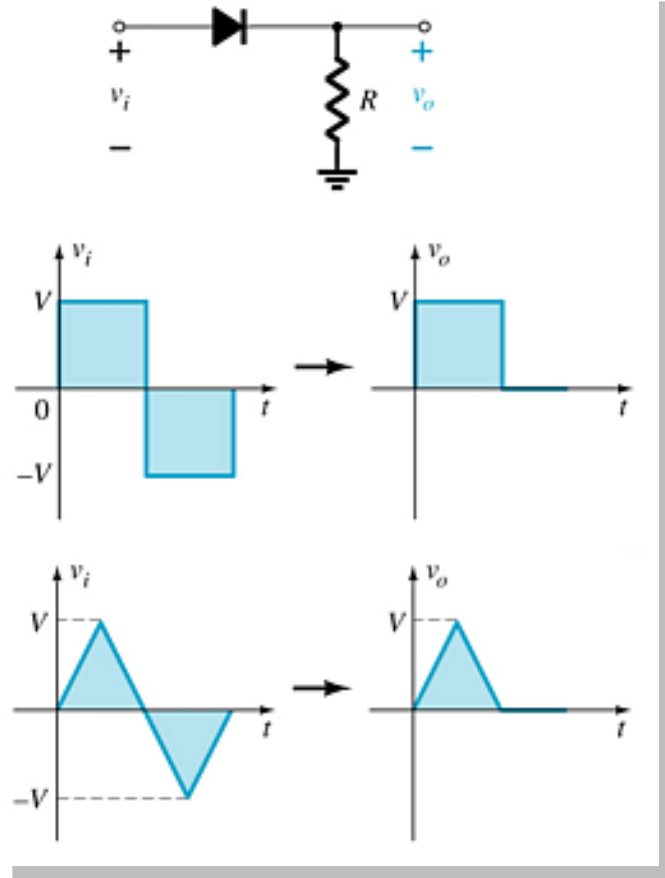
Full Wave Rectifier with Smoothing Capacitor (AC to DC Converter)



Diode Clippers

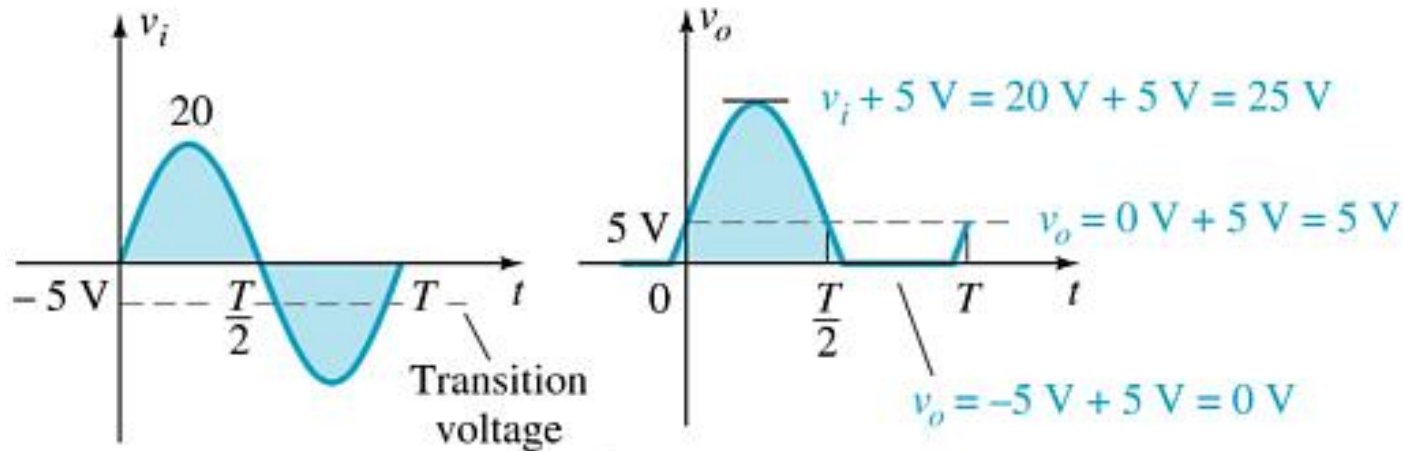
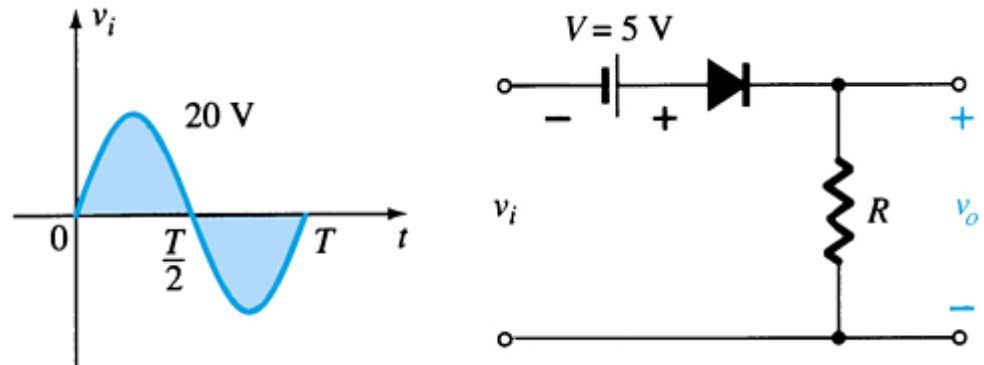
The diode in a **series clipper** “clips” any voltage that does not forward bias it:

- A reverse-biasing polarity
- A forward-biasing polarity less than 0.7 V (for a silicon diode)



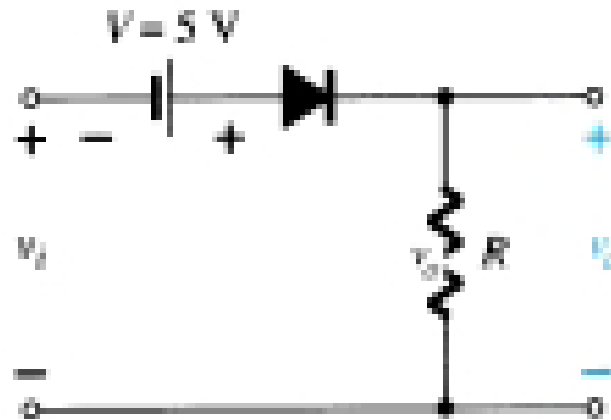
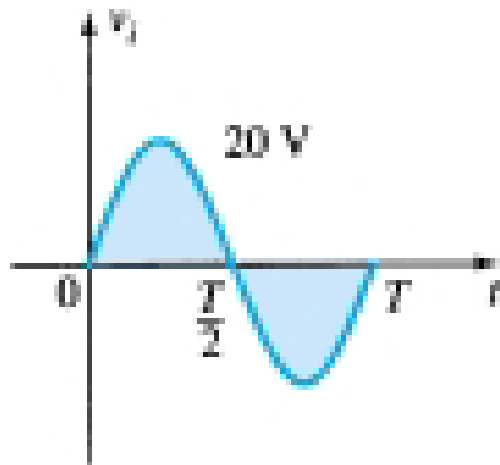
Biased Clippers

Adding a DC source in series with the clipping diode changes the effective forward bias of the diode.

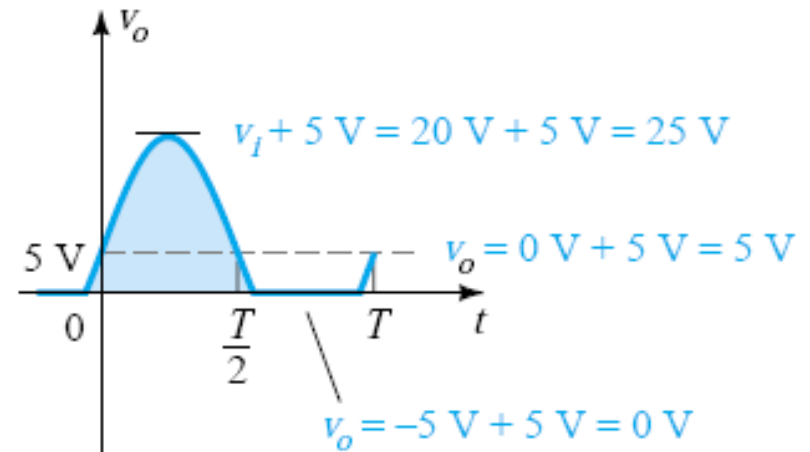
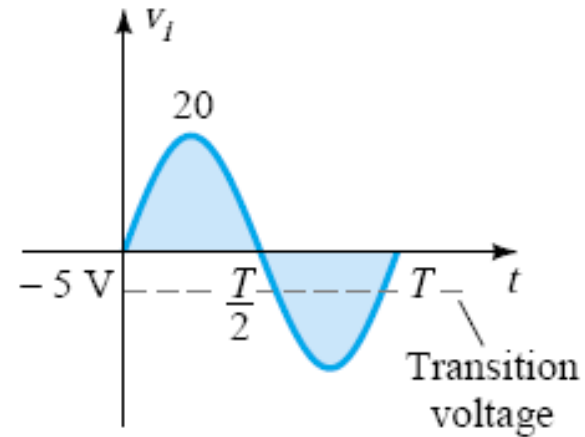
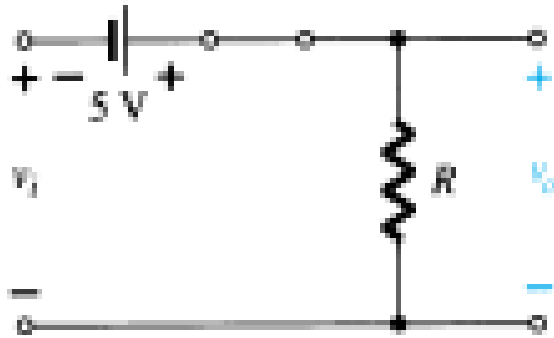


Example

Determine the output waveform for the network

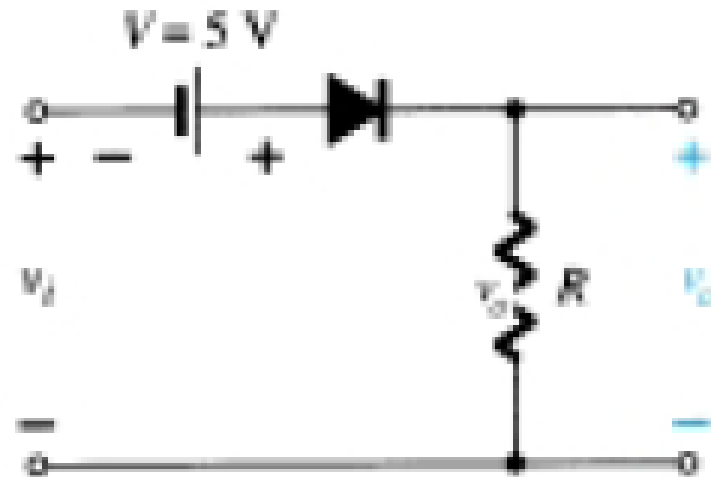
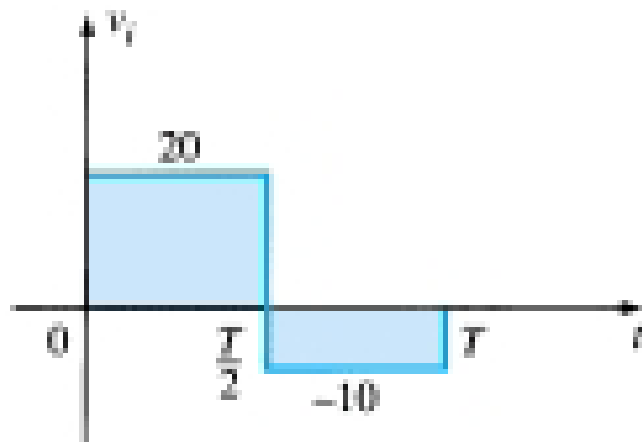


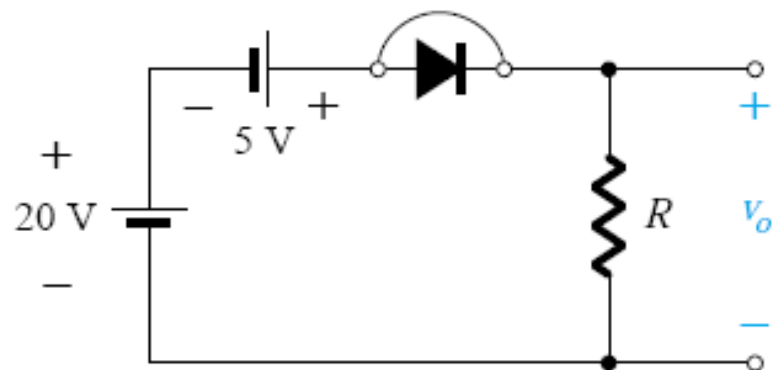
The network will appear as below, and $v_o = v_i + 5V$.



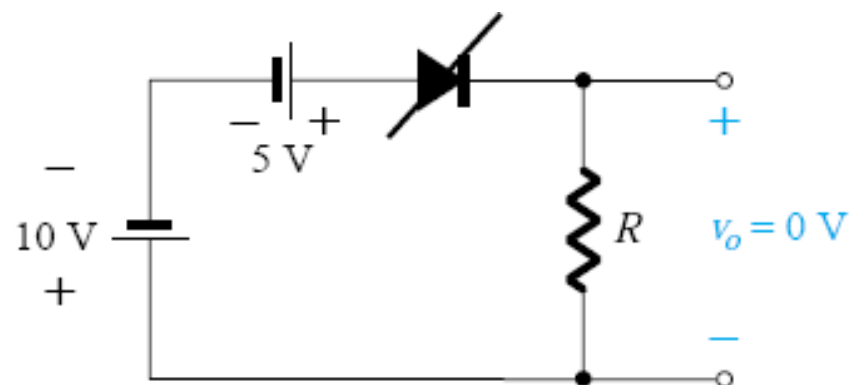
Example

Determine the output waveform for the network

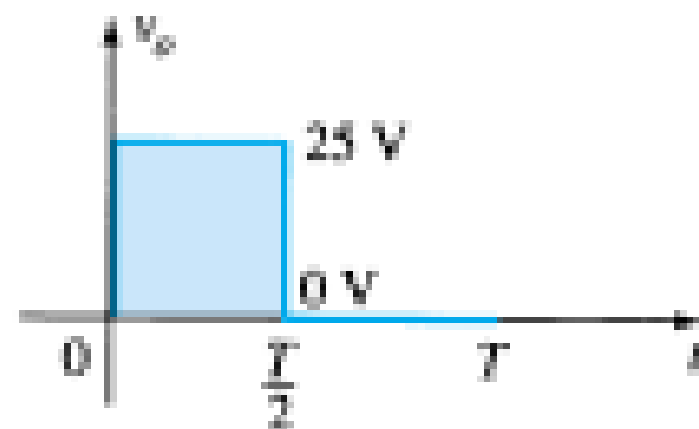




v_o at $v_i = +20$ V.



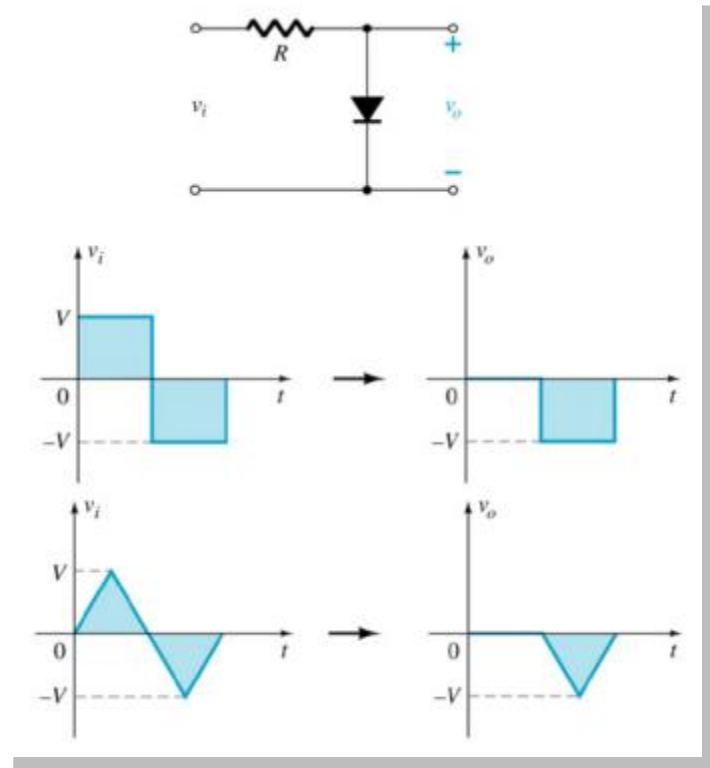
v_o at $v_i = -10$ V.



Parallel Clippers

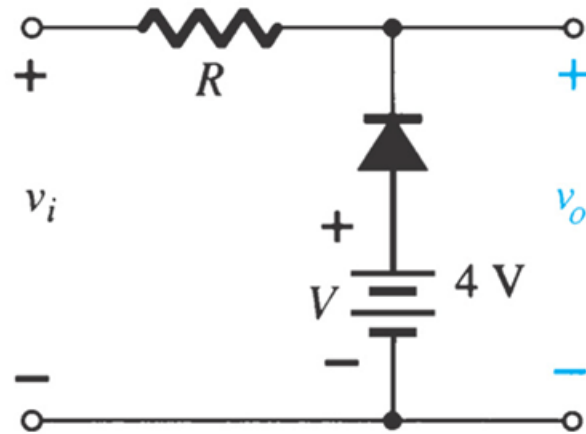
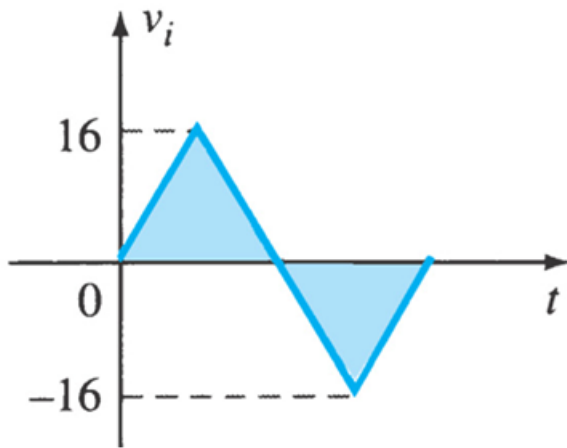
The diode (ideal) in a **parallel clipper** circuit “clips” any voltage that forward bias it.

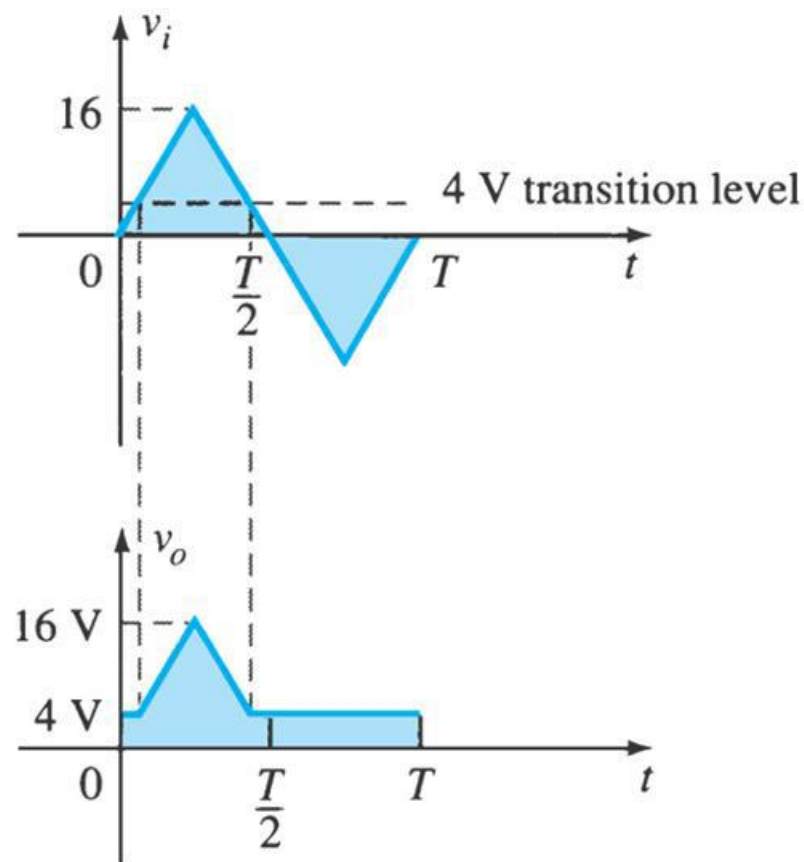
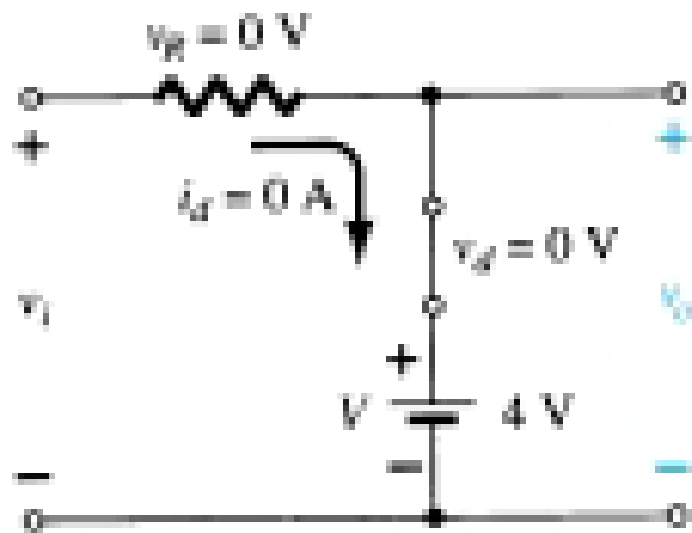
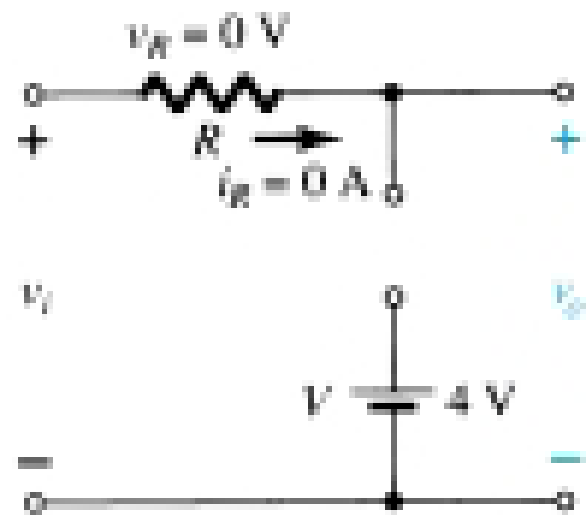
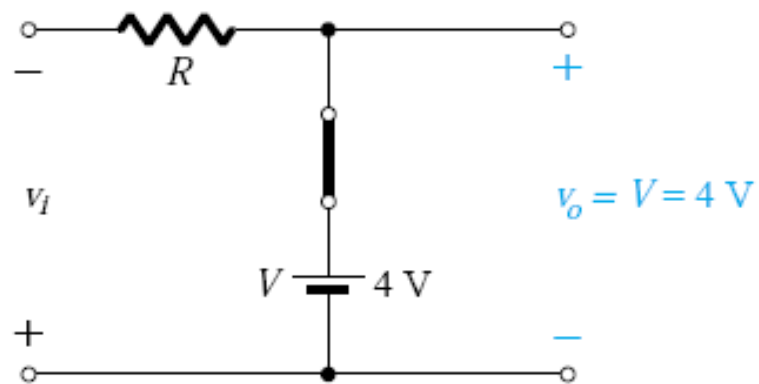
DC biasing can be added in series with the diode to change the clipping level.



Example

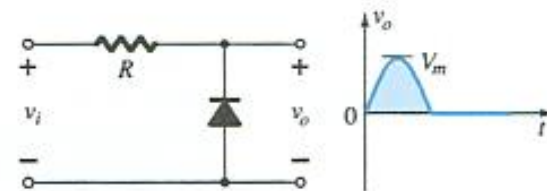
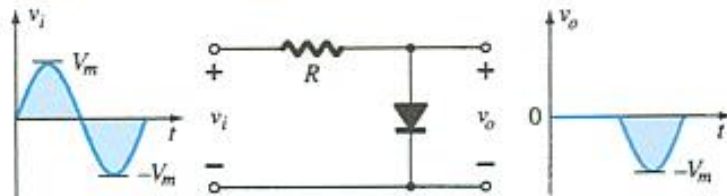
Determine v_o for the following network



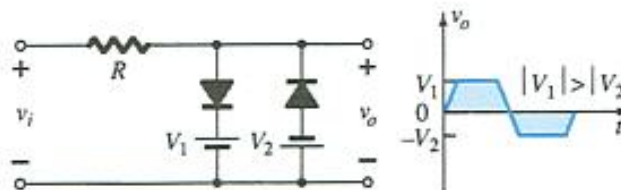
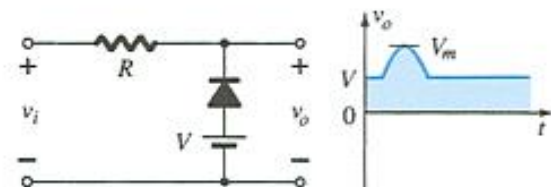
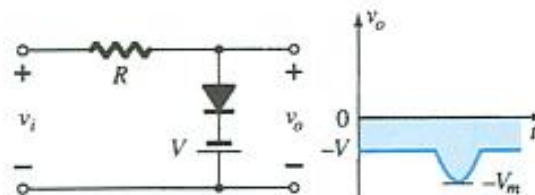
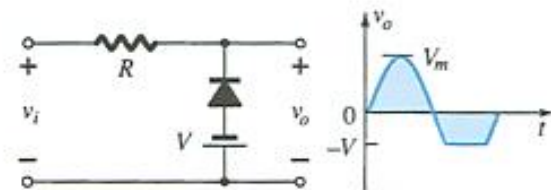
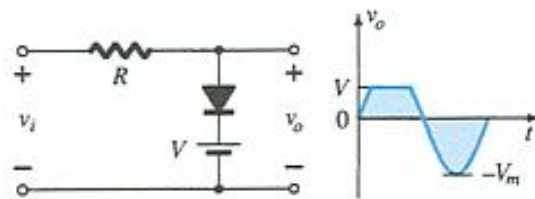


Summary of Clipper Circuits

Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)

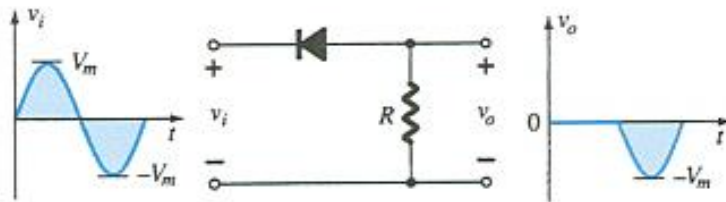


more...

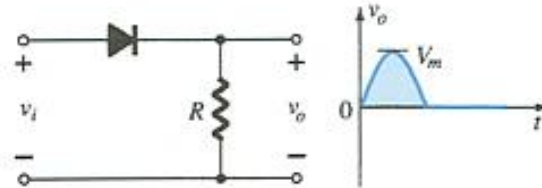
Summary of Clipper Circuits

Simple Series Clippers (Ideal Diodes)

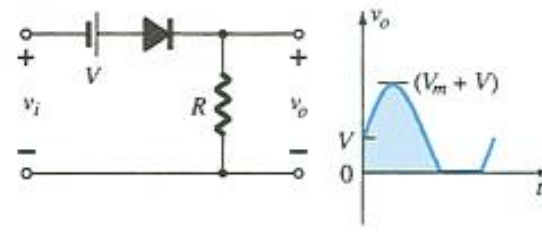
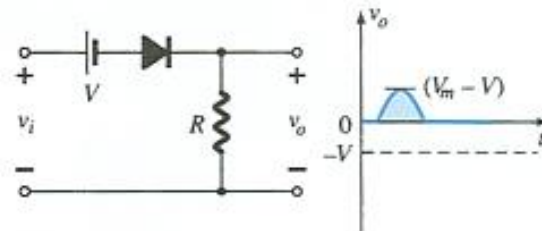
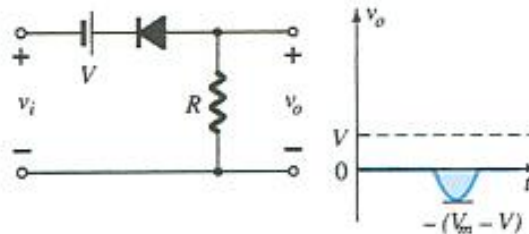
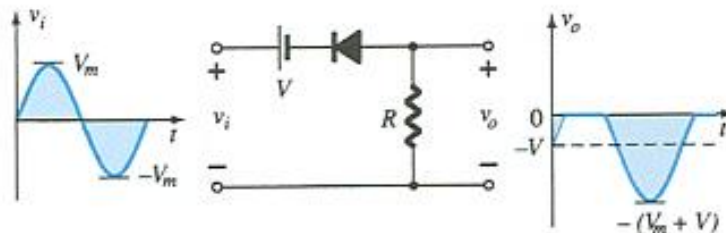
POSITIVE



NEGATIVE

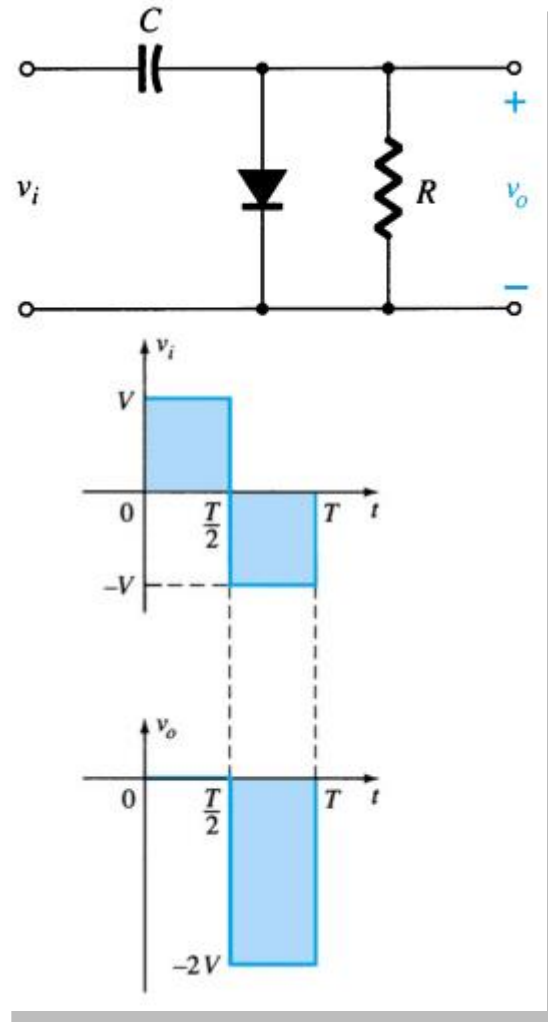


Biased Series Clippers (Ideal Diodes)



Clampers

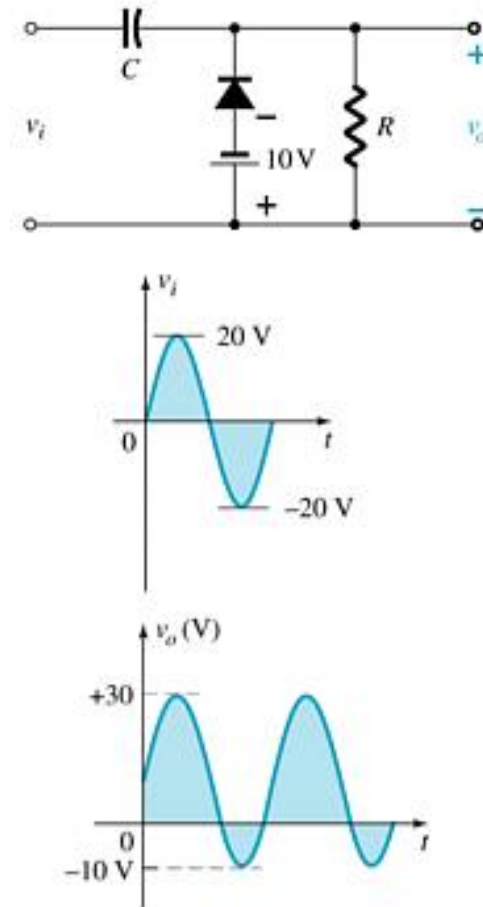
A diode and capacitor can be combined to “clamp” an AC signal to a specific DC level.



Biased Clamper Circuits

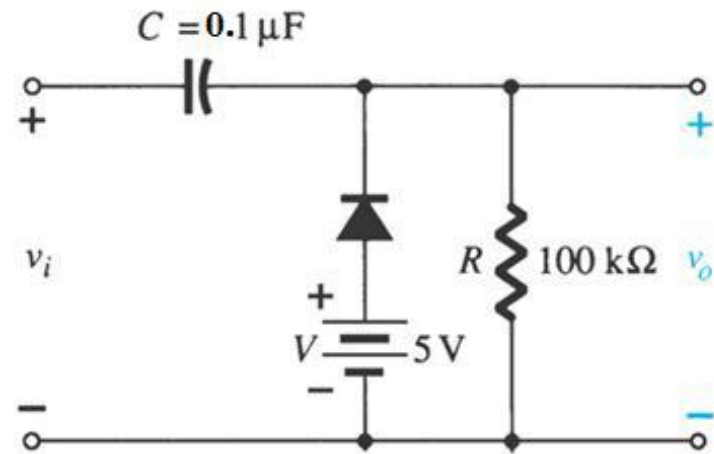
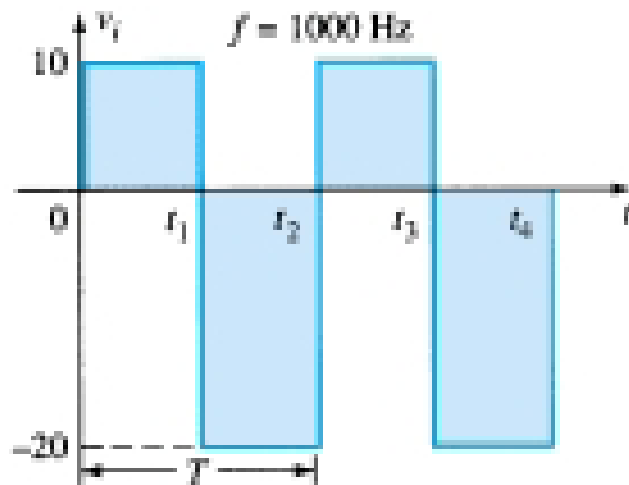
The input signal can be any type of waveform such as sine, square, and triangle waves.

The DC source lets you adjust the DC clamping level.



Example

Determine v_o for the following network for the input indicated.

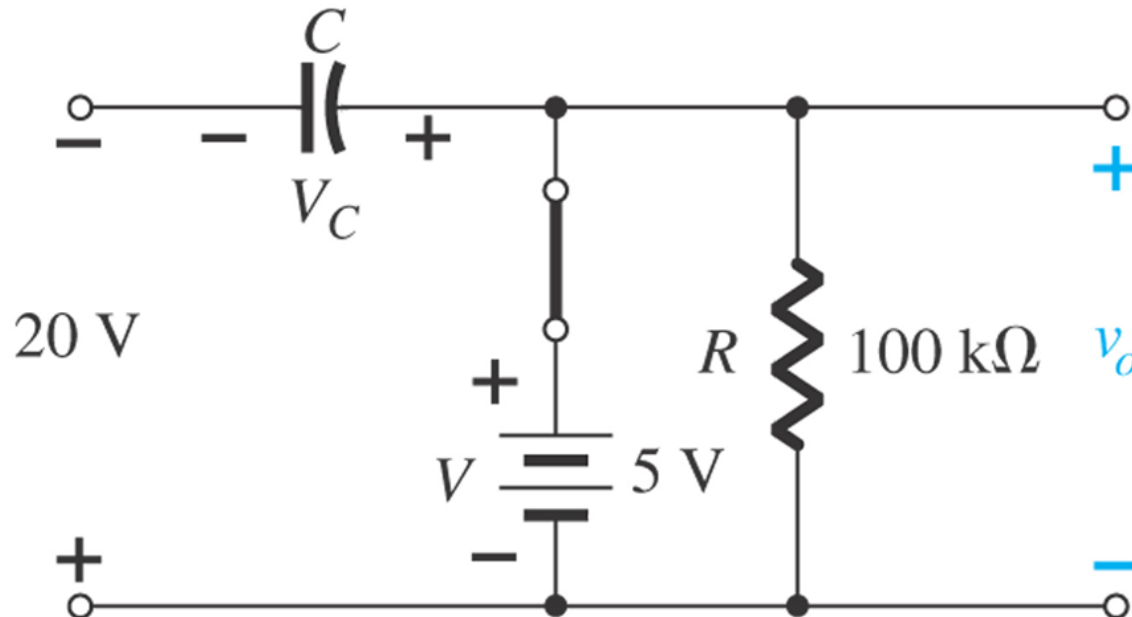


- The analysis will begin with the period $t1 \rightarrow t2$ of the input signal since the diode is in its short-circuit state as recommended by comment 1.
- For this interval the output is across R , but it is also directly across the 5-V battery the result is $V_o = 5\text{ V}$ for this interval.
- Applying Kirchhoff's voltage law around the input loop will result in

$$-20\text{ V} + V_C - 5\text{ V} = 0$$

$$V_C = 25\text{ V}$$

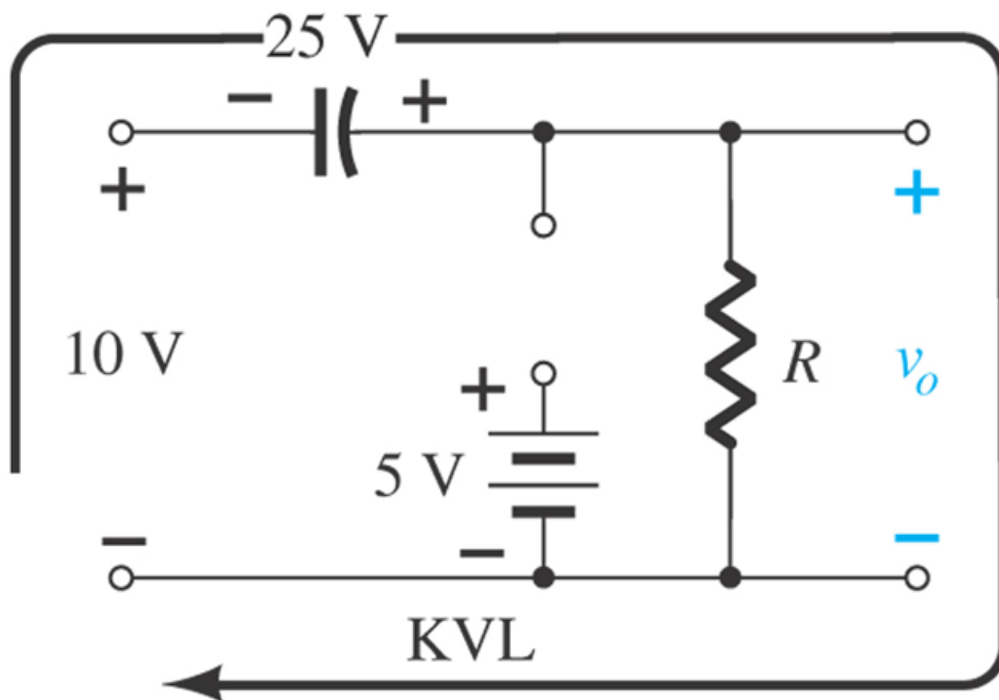
- The capacitor will therefore charge up to 25 V, as stated in comment 2.

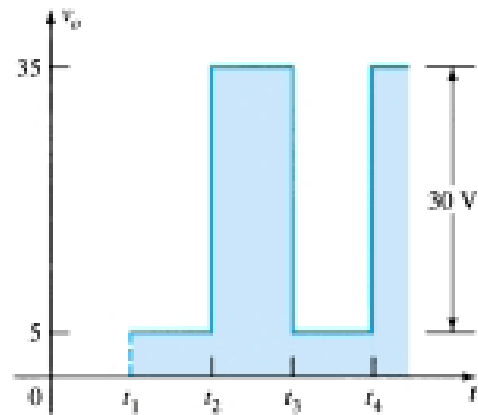
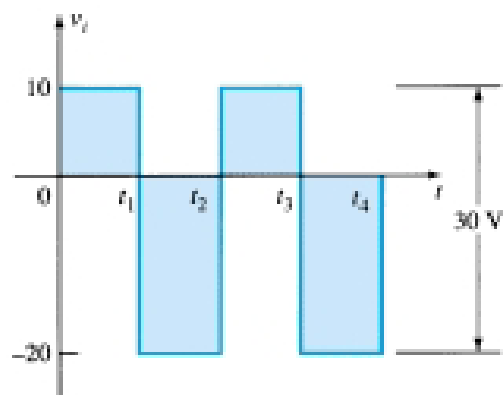
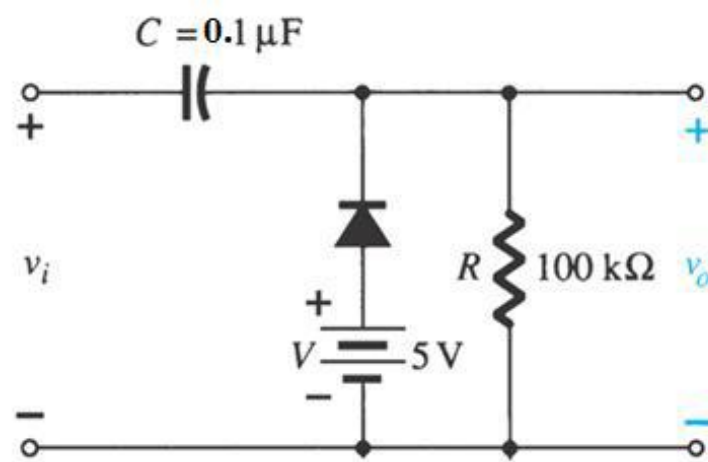


- The open-circuit equivalent for the diode will remove the 5-V battery from having any effect on V_o , and applying Kirchhoff's voltage law around the outside loop of the network will result in

$$+10\text{ V} + 25\text{ V} - v_o = 0$$

$$v_o = 35\text{ V}$$





Summary of Clamper Circuits

Clamping Networks

